test	run	pulses L / R	tape fwd, side	result L / R
roll	dropped, tape	9814 / 9755	197.0, -	0.02007 / 0.02019
roll	dropped, tape	9905 / 9842	195.0, 8.0	0.01973 / 0.01977
roll	1	9917 / 9851	201.0, 17.0	0.02036 / 0.02031
roll	2	9774 / 9778	200.0, 3.0	0.02048 / 0.02044
roll	3	9871 / 9854	201.0, 5.0	0.02039 / 0.02037
roll	**mean 1-3**			**0.02041 / 0.02037**
roll	old		15.08 / 693	0.02176
circle	left	7995 / 16091	2 turns	13.10 cm
circle	left	7979 / 16056	2 turns	13.07 cm
circle	left	8066 / 16235	2 turns	13.22 cm
circle	right	16218 / 8057	2 turns	13.28 cm
circle	right	16255 / 8059	2 turns	13.34 cm
circle	right	16015 / 7931	2 turns	13.16 cm
circle	**mean**			**13.19 cm**
circle	ruler			10.8 cm
check	1, terminal	4940 / 4940	100, -2	100.7 cm, +0.71 %
check	2	4933 / 4927	100, -1.5	100.5 cm, +0.52 %
check	3, typo fixed	4935 / 4949	100, -2	100.8 cm, +0.76 %

Exhibit 11A. Calibration of my rover, calibrate.py, 29 Oct 2026, steps 1 to 3. Roll: about 2 m at 30 rpm, tape measure beside the path. The two dropped rolls hit the tape mark and turned, so the wheels slipped. Circle: 20 / 40 rpm, 2 turns. The effective track is 22 percent above the ruler: the tyres slip sideways in a turn. Check: 100 cm by encoder against the tape. All three read 0.5 to 0.8 percent far, under 8 mm.